

Pollinating *Amorphophallus*

The female flowers open on **day one** and are receptive for **hours**. The plant's own pollen does not exist until day two, by which time those stigmas have shut. So the cross is never limited by technique — a brush on a sticky stigma is not a hard skill. It is limited by whether there is **viable pollen in the building on the one morning it matters**. So collect from every inflorescence you get: the freezer is the plan, not the backup.

THE SEQUENCE

2–4 MONTHS	Bud pushes from the dormant corm. Squeeze it: soft and round-tipped = inflorescence; hard and pointed = leaf. ⁶
3–5 WEEKS	Bud to full bloom. Cataphylls dry off in order; the appendix forms last. ⁶
DAY 1	Spathe fully open · scent at full strength · stigmas wet. Pollinate now.
DAY 1→2	Appendix swells, then shrinks below its starting size . Deflating = the window has shut. ⁵
DAY 2–3	Anthers open; pollen appears as dust, paste or strands. Collect. Do not try to pollinate — the stigmas below are finished.
3–13 DAYS	Spathe withers, collapses, peduncle detaches. Normal — watch the base. ⁶
2–4 MONTHS	Berries ripen. Harvest when soft . ²

THREE NUMBERS

<12 h	The pistillate phase in most species. ¹ No second chance on the same inflorescence.
17 mo	Longest gap on record between collecting pollen and a cross that took, frozen at −24 °C. ² Shortest success: 11 days .
90–95%	Viability of healthy fresh pollen. ³ A stunted male zone shedding almost nothing is a bad donor, visibly.

KIT

■ Fine brush	■ Desiccant (silica gel)
■ Scalpel or blade	■ Freezer, −24 °C
■ Gloves — raphides float free in the stigmatic fluid ⁴	■ Dark pen, to mark zones
■ Glass tube, ~5 × 35 mm	■ Labels + dated notebook

KEEPING POLLEN — THE PART THAT DECIDES IT

- 01 **Collect the moment the anthers open.** Pollen sitting in an open spathe in a warm room loses something every hour.
- 02 Into a **small, dry glass tube with a desiccant**. A big container holds a big volume of humid air against the pollen — the one published programme used none and sometimes found it wet.²
- 03 **Date the label.** The storage record in this genus survives only because every tube carried one.
- 04 **Freeze at −24 °C.** Every success beyond a month came from a freezer.²
- 05 **Thaw sealed, one hour, then use.** Never open it cold — that condenses water onto the pollen.

Two plants bought separately may be one clone — this genus spreads by offsets and leaf bulbils. Get pollen from a different *collection*, not a different pot.

ON THE DAY — ALL THREE, NOT ONE

Fully open The limb has finished unfurling. A spathe still parting is not yet in the female phase.	Full scent Peak intensity and receptivity coincide. Pollinate <i>after</i> the scent develops.	Stigmas wet Visible fluid, tacky to a brush. Dry means you are early or late.
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THE CROSS

- 01 Confirm all three checks above. **Day one, as early as possible** — that is the published instruction and there is no softer version of it.
- 02 **Cut a window in the front of the spathe only**, about as much as you would open to photograph the interior. **Cut low:** the female zone is at the *base*, inside the closed part — not at the top, whatever you may have read.
- 03 **Leave the cut open.** No bag, no foil, no tape. The stigmas are “*very sticky and no pollen will fall off*”, so there is nothing for a bag to retain — and sealing a warm wet chamber rots it.²
- 04 If the pollen came from the freezer, **thaw it sealed** for about an hour at room temperature. Opening a cold tube in a warm room condenses water straight onto it.²
- 05 **Brush every stigma you can reach.** One application. In a species whose female phase runs under twelve hours there is rarely a second morning to repeat on.
- 06 Using more than one donor? **Mark the spadix into zones** with dashes in dark ink before you start. One inflorescence can then test several sources — which matters when the next one is three years away.²
- 07 **You will not reach the back.** Nobody does. The fruit head will ripen lopsided and *bend* toward the side that set. Documented, and harmless.²
- 08 **Label it and stop touching it.** Date, seed parent, pollen parent, pollen age.

WHEN NOTHING HAPPENS

NOTHING SWELLED	Wrong day, almost always. Judge by the stigmas , not the calendar. Dead pollen and an incompatible cross look identical from outside.
A FEW, THEN STOP	Too few stigmas reached. Cover more of the zone next time; consider a second donor on marked zones.
HEAD BENT	Normal. You could not reach the rear stigmas. It still ripens.
ROTTED OFF ON TOP	Normal. That is the end of anthesis. A pollinated base keeps developing under it.
MONTHS OF NOTHING	Still coming. Fresh seed of one species took 7–11 months to germinate. ² Do not discard a pot for a year.
NO LEAF AFTERWARDS	Expected if it <i>fruited</i> — Asian species skip the season. Feed the corm the year before . A plant that fails to fruit leafs normally.

Colour is not a ripeness rule. Most species ripen green to orange-red, but *Amorphophallus yuloensis* ripens **violet blue**.⁵ Know your species' fruit colour before you start waiting — otherwise go by **softness**, which is the cue that actually produced seed. Then de-pulp, wash, and sow ~1 cm deep in half sand, half potting mix at 25–28 °C. **Never let the seed dry:** once dry it cannot be brought back.

SOURCES ¹ Claudel et al. 2023, *Plant J.* 115: 874–894 · ² Kiehlmann & Kiehlmann 2003, *Aroideana* 26: 56–65 · ³ Anil et al. 2023, *Genet. Resour. Crop Evol.* · ⁴ Raman et al. 2017, *Turk. J. Bot.* 41: 63–74 · ⁵ Barnett 2024, *Aroideana* 47(3): 15–24 and Tran et al. 2017, *Aroideana* 40(1): 57–62 · ⁶ Handayani et al. 2020, *Biodiversitas* 21: 5835–5844. — Every figure is scoped to the species it was measured in.